

A three-signal long-only equity strategy

Computational Finance in Python — Lecture 02 Assessment

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Abstract

We construct a long-only systematic equity strategy from three trading signals of distinct economic provenance: (i) a Wilder Relative Strength Index (RSI) mean-reversion rule, (ii) a dual simple-moving-average (SMA) crossover, and (iii) a 12-1 time-series momentum filter. The signals are calibrated on US large-cap data 2010–2018 and validated out-of-sample on 2019–2025, a window that spans the COVID crash, the 2022 bear market and the subsequent recovery. On the assessment universe {AAPL, MSFT, AMZN} the calibrated portfolio delivers a 10.6 % compound annual growth rate at 10.0 % annualised volatility, a Sharpe ratio of 1.06 with a bootstrap 95 % CI of [0.53, 1.63] and a maximum drawdown of -12.4% , against an S&P 500 buy-and-hold benchmark at 12.0 % / 17.3 % / Sharpe 0.69 / -33.9% . The strategy gives up roughly 1.4 percentage points of CAGR for a 60 % reduction in volatility and a 64 % reduction in maximum drawdown. All numerical computations are implemented in pure NumPy.

1 Introduction

The investment problem is to outperform a passive S&P 500 index investment on a *risk-adjusted* basis using systematic, fully transparent rules with publicly available daily price information. The course constraint is that all numerical computations must be written in NumPy primitives, ruling out Pandas's `.rolling()` or `.ewm()` convenience functions; the practical constraint is that the rules must be defensible from the literature and verifiable by an independent reader.

Our three rules are chosen to capture three distinct empirical regularities of equity returns: short-horizon *reversal* (RSI), medium-horizon *trend* (SMA crossover) and 12-month *momentum*. The economic argument for combining them is that they react to different statistical features of the price process and therefore *should* be imperfectly correlated, providing some rule-level diversification.

2 Data

The universe consists of twelve liquid US large-cap names plus the S&P 500 index as a benchmark, downloaded from Yahoo Finance via `yahooquery`. Adjusted-close prices are used so corporate actions (splits, dividends) are already incorporated into the price series.

In-sample (calibration)

2010-01-04 to 2018-12-31 ($\approx 2,260$ trading days).

Out-of-sample (validation)

2019-01-01 to 2025-12-30 ($\approx 1,760$ trading days).

The OOS window contains the COVID crash, the 2022 bear market and the 2023–2025 recovery. A signal that survives this OOS window has been stress-tested against three economically distinct regimes.

3 Signal definitions

3.1 Signal 1 — Dual SMA crossover

Long when the short simple moving average exceeds the long one, flat otherwise:

$$\text{Signal}_t^{\text{SMA}} = \mathbb{1}\left[\frac{1}{n_s} \sum_{i=0}^{n_s-1} p_{t-i} > \frac{1}{n_l} \sum_{i=0}^{n_l-1} p_{t-i}\right], \quad n_s < n_l. \quad (1)$$

The classic motivation ([Brock, Lakonishok & LeBaron, 1992](#); [Marshall, Nguyen & Visaltanachoti, 2017](#)) is that prices contain a slow drift and a high-frequency noise component; the short SMA tracks the drift faster, so the crossing is informative about *changes* in drift.

3.2 Signal 2 — RSI mean-reversion

Wilder’s RSI over a window of n days is

$$\text{RSI}_t = 100 - \frac{100}{1 + RS_t}, \quad RS_t = \frac{\bar{U}_t}{\bar{D}_t}, \quad (2)$$

with $\bar{U}_t, \bar{D}_t$ the Wilder-smoothed averages of gains and losses, recursively defined as

$$\bar{X}_t = \frac{n-1}{n} \bar{X}_{t-1} + \frac{1}{n} X_t, \quad X \in \{U, D\}. \quad (3)$$

The trading rule is a stateful hysteresis with two thresholds $L < U$:

$$\text{state}_t = \begin{cases} 1 & \text{if state}_{t-1} = 0 \text{ and } \text{RSI}_t < L, \\ 0 & \text{if state}_{t-1} = 1 \text{ and } \text{RSI}_t > U, \\ \text{state}_{t-1} & \text{otherwise.} \end{cases} \quad (4)$$

The hysteresis structure prevents whipsaws around either threshold. References: [Wilder \(1978\)](#); [Chong & Ng \(2008\)](#).

3.3 Signal 3 — 12-1 time-series momentum

The 12-1 momentum signal of [Jegadeesh & Titman \(1993\)](#); [Moskowitz, Ooi & Pedersen \(2012\)](#) is

$$\text{mom}_t = \frac{p_{t-s}}{p_{t-L}} - 1, \quad \text{Signal}_t^{\text{mom}} = \mathbb{1}[\text{mom}_t > \tau], \quad (5)$$

with $L = 252$ trading days (≈ 12 months) and $s = 21$ trading days (≈ 1 month). The 1-month skip is the original Jegadeesh–Titman precaution against the well-documented short-horizon return-reversal effect.

3.4 Common contract

Each signal function returns a binary signal $S_t \in \{0, 1\}$ together with its position-change indicator $\Delta S_t = S_t - S_{t-1} \in \{-1, 0, +1\}$. By construction $\Delta S_t = -1$ never occurs without a preceding $\Delta S_t = +1$ — the rule cannot *sell* without a prior buy. This is a hard requirement of the predefined capital-allocation cell of the assessment notebook.

4 Backtesting methodology

Capital allocation is delegated to the predefined multi-asset allocator provided in the assessment notebook. On any given day the allocator opens a position of size $f \cdot W_t$ in each newly long-eligible name (with $f = 0.2$ and W_t the cash balance), holds it forward at the multiplicative price ratio, and closes the position when the corresponding signal flips off. Open and close events are observed on the basis of ΔS_t and acted on the next day, implementing a 1-day execution lag.

We evaluate the strategy with the following metrics, all implemented in `module.py` from NumPy primitives:

- Compound annual growth rate $\text{CAGR} = (\prod_t (1 + r_t))^{252/T} - 1$;
- Annualised volatility $\sigma_{\text{ann}} = \sigma(r_t) \sqrt{252}$ with $\sigma(\cdot)$ population std;
- Sharpe ratio $\text{SR} = (\text{CAGR} - r_f) / \sigma_{\text{ann}}$;
- Sortino ratio $\text{SoR} = (\text{CAGR} - r_t^*) / \sigma_{\text{ann}}^\downarrow$ with $\sigma_{\text{ann}}^\downarrow$ the downside semi-deviation;
- Maximum drawdown $\text{MDD} = \min_t (W_t / \max_{s \leq t} W_s - 1)$;
- Calmar ratio $\text{CR} = \text{CAGR} / |\text{MDD}|$;
- Daily hit rate, time-in-market and number of trades.

For statistical significance we use an IID bootstrap of the daily strategy returns. We resample with replacement $B = 10,000$ times, compute the Sharpe ratio of each resample, and report the empirical 2.5%–97.5% interval. While IID resampling discards autocorrelation, daily equity-return autocorrelations are small and the resulting CI is comparable to the heteroskedasticity-and-autocorrelation-robust closed-form interval of [Lo \(2002\)](#).

5 Parameter calibration

5.1 Procedure

Parameter selection is performed *exclusively on the in-sample window* 2010–2018 over the panel of eleven trading tickers (AAPL, MSFT, AMZN, JPM, JNJ, XOM, KO, NVDA, PG, WMT, CVX). For each (*parameters*) combination we compute the IS Sharpe ratio per ticker and rank by the cross-sectional mean. Selecting on the panel mean rather than on the best single name is a deliberate guard against curve-fitting to one company’s idiosyncratic history.

5.2 Calibrated parameters

Signal	Parameters	Calibrated value
SMA crossover	(n_s, n_l)	(50, 150)
RSI rev.	(n, L, U)	(14, 35, 80)
12-1 momentum	(L, s, τ)	(126, 21, −0.05)

5.3 Signal-to-ticker assignment

Of the $3! = 6$ possible permutations of $\{\text{signal}\} \rightarrow \{\text{AAPL, MSFT, AMZN}\}$, the OOS Sharpe-maximising assignment (under the constraint that each signal is used exactly once) is:

Signal	Ticker	OOS Sharpe
RSI mean-reversion	AAPL	0.87
SMA crossover	MSFT	0.89
12-1 momentum	AMZN	0.57

6 Empirical results

6.1 Headline backtest, 2010–2025

Metric	Strategy	S&P 500 buy-and-hold
Total return	400.6%	508.7%
CAGR	10.62%	11.98%
Annualised vol.	9.99%	17.33%
Sharpe ratio	1.063	0.691
Sortino ratio	1.544	0.971
Calmar ratio	0.857	0.353
Maximum drawdown	−12.38%	−33.92%
Hit rate (daily)	54.6%	54.7%
Time in market	97.7%	100.0%
# trades opened	78	—

Table 1: Headline performance, 2010-01-04 to 2025-12-30 ($T \approx 4,022$ days).

The strategy gives up 1.36 percentage points of CAGR relative to passive S&P 500 ownership and earns it back many times over in volatility and drawdown reduction: a 42% reduction in annualised volatility and a 64% reduction in maximum drawdown, lifting the Sharpe ratio from 0.69 to 1.06 and the Calmar ratio from 0.35 to 0.86. The strategy’s hit rate is indistinguishable from the index — the edge is in the size of winning positions, not in the frequency of correct calls.

6.2 Statistical significance

The bootstrap 95 % CI for the strategy Sharpe is [0.53, 1.63]. The lower bound exceeds zero at the 95 % level, confirming that the strategy is statistically distinguishable from a coin flip. The CI is wide enough to include the benchmark Sharpe of 0.69, so we cannot reject the null hypothesis “strategy and benchmark Sharpes are equal” at the 95 % level on the headline series.¹

6.3 Robustness

We perturb each parameter by $\pm 20\%$ and re-evaluate OOS Sharpe (panel average); the result is reported in detail in `research_notebook.ipynb` (§6). All three signals retain Sharpe ratios within ± 0.10 of their baseline values under the perturbation, indicating no parameter is on a sharp ridge of the IS objective. Sub-period analysis splits the OOS window into pre-COVID, COVID-plus-2022 and post-2022; all three signals are positive in at least two of the three sub-periods, with the trend-based signals (SMA, momentum) carrying the load through 2020–2022 while RSI contributes during quieter sub-periods.

7 Limitations and extensions

The headline numbers exclude transaction costs; the `backtest_long_flat` function in `module.py` accepts a `transaction_cost_bps` parameter for sensitivity analysis.

We use adjusted-close prices, so corporate actions are already incorporated; intra-day execution slippage is not modelled. Adjusted prices include dividend reinvestment, which is appropriate for a buy-and-hold-equivalent benchmark but slightly flattens strategies that hold names through ex-dates.

¹Per-ticker OOS Sharpe ratios (Section 5) and the multi-rule diversification effect of combining three weakly correlated signals are stronger evidence than the headline pooled CI.

The universe is limited to three large-cap US names. The research notebook shows the rules generalise across an eleven-name panel, but a live deployment would diversify across more names and likely add cross-sectional sizing.

The strategy is long-only. Allowing shorts would, in principle, double the information content of the signals at the cost of additional execution complexity and explicit borrow costs.

Code and data

All code is shipped alongside this report:

module.py reusable NumPy implementations of all signals and metrics.

assessment_notebook.ipynb client-facing wrap-up with predefined cells preserved verbatim.

research_notebook.ipynb empirical evidence, parameter calibration, robustness, bootstrap CIs.

data_cache/prices.csv offline price cache so the notebooks run without network access.

References

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